

Simulation of off-grid generation options for remote villages in Cameroon

E.M. Nfah^{a,*}, J.M. Ngundam^b, M. Vandenberg^c, J. Schmid^c

^a*I.U.T. Fotso Victor, University of Dschang, P.O. Box 134, Bandjoun, Cameroon*

^b*School of Engineering, University of Yaounde I, P.O. Box 8390, Cameroon*

^c*ISET e.V., Königstor 59, D-34119 Kassel, Germany*

Received 10 July 2006; accepted 19 May 2007

Available online 17 August 2007

Abstract

Off-grid generation options have been simulated for remote villages in Cameroon using a load of 110 kWh/day and 12 kWp. The energy costs of proposed options were simulated using HOMER, a typical village load profile, the solar resource of Garoua and the flow of river Mungo. For a 40% increase in the cost of imported power system components, the cost of energy was found to be 0.296 €/kWh for a micro-hydro hybrid system comprising a 14 kW micro-hydro generator, a 15 kW LPG generator and 36 kWh of battery storage. The cost of energy for photovoltaic (PV) hybrid systems made up of an 18 kWp PV generator, a 15 kW LPG generator and 72 kWh of battery storage was also found to be 0.576 €/kWh for remote petrol price of 1 €/l and LPG price of 0.70 €/m³. The micro-hydro hybrid system proved to be the cheapest option for villages located in the southern parts of Cameroon with a flow rate of at least 200 l/s, while the PV hybrid system was the cheapest option for villages in the northern parts of Cameroon with an insolation level of at least 5.55 kWh/m²/day. For a single-wire grid extension cost of 5000 €/km, operation and maintenance costs of 125 €/yr/km and a local grid power price of 0.1 €/kWh, the breakeven grid extension distances were found to be 15.4 km for micro-hydro/LPG generator systems and 37.4 km for PV/LPG generator systems respectively. These results could be used in Cameroon's National Energy Action Plan for the provision of energy services in the key sectors involved in the fight against poverty.

© 2007 Elsevier Ltd. All rights reserved.

Keywords: Rural electrification; Off-grid generation options; Cost of energy; Breakeven grid distance

1. Introduction

Situated in central Africa, Cameroon lies between latitudes 1°40' and 13°05' to the north of the equator and longitudes 8°30 and 16°10 to the East of longitude 0°. The country is organised into 10 provinces and each province is further structured into divisions, sub-divisions, districts and sub-districts as administrative units. The population of Cameroon in 2001 was estimated at 15.5 million inhabitants unequally distributed on a surface area of 475,440 km² [1].

In terms of energy resources, Cameroon has the second hydroelectric potential (294 TWh) in Africa after the

Democratic Republic of Congo (about 1000 TWh) [2]. In 2002, the installed hydroelectric and thermal generation capacity in Cameroon was 847 MW of which hydroelectricity accounted for 85% and thermal electricity 15%. There are three independent grids in the South, East and North of the country. The southern grid is supported by two hydroelectric stations at Songloulou and Edea and four thermal plants in Limbe, Douala, Yaoundé and Bafoussam. The northern grid is supported by a hydroelectric station at Lagdo and a thermal plant at Garoua. The eastern grid is supported by a unique thermal plant at Bertoua [2]. These grids supply electricity mainly to provincial and divisional capitals, while most sub-divisional and district capitals run on independent mini-grids supported by thermal stations. The locations of existing grids in the South, East and North of the country are indicated by the shaded areas of Fig. 1.

*Corresponding author. Tel.: +237 9688 2160; fax: +237 3344 2449.

E-mail address: em_nfah@yahoo.fr (E.M. Nfah).

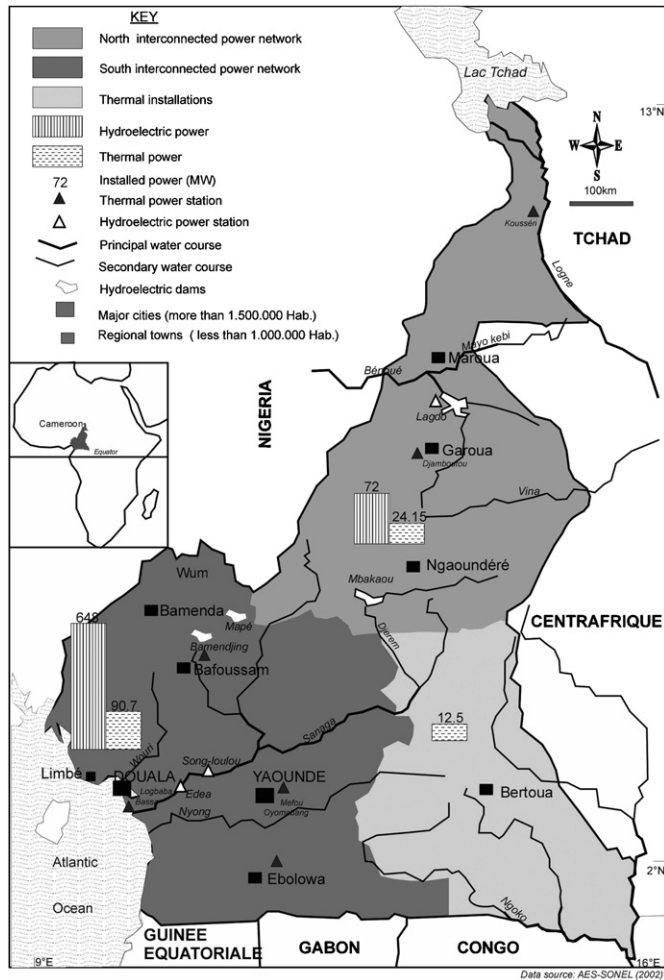


Fig. 1. Map of electricity generation and distribution networks in Cameroon in 2002.

In spite of the substantial availability of hydroelectric resources in Cameroon, electricity supply remains a serious problem as evidenced by frequent power cuts during the months of January to June since 1999 [2,3]. Power cuts affect the activities of industries, administrative and social services as well as those of urban and rural households with utility connections. Consequently, the total access rate to electricity in Cameroon is quite low and has been estimated at 11% [3]. This low access rate is partly due to the slow rate of exploitation of the hydroelectric potential of the country and partly due to the slow rate of expansion and/or upgrading of existing grids to match growing energy needs. To redress the energy crisis specifically in the southern interconnected grid where low water volumes in dams during the dry season affect hydroelectric production [3], the electricity power corporation and the government have embarked on hydrothermal expansion. This approach has been proposed in [4]. However, optimal scheduling of large-scale hydrothermal power systems that has been proposed in [5] is yet to be implemented. In spite of the expansion of thermal generation capacities and future development of hydroelectric installations,

many villages will still remain without electricity for many years.

In Cameroon, villages at the periphery of sub-urban grid connected areas are often electrified through the extension of medium voltage 30-kV single-wire lines over distances that are often less than 5 km, the installation of auto-transformers in the power range 10–50 kVA and a mini-grid supplying power in a radius in the range 500–1000 m. This practice has produced low electrification rates estimated to be in the range 4–6% for sub-urban and rural areas that harbour 62–67% of the population [2]. The electrification rate in rural areas is limited by the absence of small-scale decentralised options that could meet the low energy needs of remote villages that often lack income to pay for grid connected service.

Households in most remote villages use electricity mainly for lighting, and to power radios and small black and white television sets. These DC loads can be met in a village by a centralised battery charging station, with households owning batteries or individual solar home systems [6,7]. These household energy needs have been found to be in the range 0.3–0.6 kWh/day in Inner Mongolia Autonomous Region (IMAR) of China and 0.5–0.6 kWh/day in Mekong countries. When a small refrigerator is included, the daily energy needs are raised to 1.2–1.6 kWh/day for households in IMAR and to 1.0–1.5 kWh/day for households in Mekong countries [8,9]. An energy survey in Cameroon has also established low energy needs in the range 0.2–1 kWh/day for lighting and powering of a radio and a television set [10]. In Nigeria [11], the proposed household load (300–340 W) in a decentralised power supply option has daily energy needs in the range 1.5–1.66 kWh. The household load consists of nine lights each rated 20 W, a radio/cassette rated 60 W and a television set rated 60 or 100 W [11]. The use of energy efficient lamps in the proposed household load can greatly reduce the power demand of each rural household. In Kenya, two community pico-hydro schemes with rated outputs of 1.1 and 2.2 kW have electrified 65 and 110 households, respectively, with light packages of 10 W. One light package is enough power for one 8 W compact fluorescent lamp (CFL) and a radio [12].

Small-scale off-grid generation options that exploit local renewable energy resources such as hydro, biomass (animal and/or human waste), solar and or wind energy can be used for the low energy needs of most remote villages. Research efforts for the promotion of renewable energy options in Cameroon include the feasibility of wind energy conversion systems for hypothetical villages requiring 70 kWh/day in northern Cameroon [13] and the modelling of solar/diesel/battery hybrid systems for remote households, clinics and a school [10]. In spite of these research efforts, the economic performance of feasible off-grid generation options for village power systems below 50 kW have not been evaluated to encourage the use of small off-grid renewable energy systems in the rural electrification policy of Cameroon.

The objective of this paper is to simulate off-grid generation options that could be used in the provision of energy services for key sectors involved in the fight against poverty reduction particularly in remote communities. Remote area power supply (RAPS) systems based on renewable energy technologies are expected to be comparatively cheaper than diesel stand alone systems and/or single-wire grid-extended systems for remote villages that require power supplies in the 10–50 kW power range.

2. Remote area power supply system models

There are three main models on which RAPS systems are based: namely, the DC, AC/DC and AC-bus configurations [14,15]. The comparison of the performance of these bus configurations has been done by Ross [15]. The DC-bus configuration is used in Solar Home Systems that are often limited to consumptive loads (e.g. light, radio, television). In order to operate productive loads (e.g. small power tools, water pumps, and refrigerators), the AC/DC or the AC-bus configuration can be used to provide power to AC-loads. According to the results obtained in [15], the AC/DC bus-configuration is more energy efficient than the AC or DC bus systems if intelligent genset dispatch strategies that limit the time that the fuel generator operates at part load are implemented. However, the AC/DC systems are not usually extendable and show a complicated DC system design and therefore higher system costs [14]. The extendable AC-bus approach in spite of being completely new has been successfully deployed in 100 systems in 14 different countries distributed in four continents [16]. The AC-bus configuration can be centralised with all power generators connected to the grid through one AC-bus, or distributed with power generators connected unto a mini-grid at multiple points.

Off-grid generation options proposed in this paper are those based on the AC-coupled modular expandable hybrid power system model that has been used in the evaluation of photovoltaic (PV) hybrid systems for remote villages in Mekong countries [9]. This model is characterized by a stipulated energy coupling (AC-bus bar with 230/400 V for example, 50 Hz with power supplied to users

over a mini-grid), a standardized information exchange and a supervisory control. This approach allows the setting up of an adaptable and expandable system structure, thus covering almost every supply situation; but it means that the generators integrated into the system have to be equipped with special control features [17]. Fig. 2 shows a three-phase configuration of an AC-coupled modular expandable hybrid power system [9]. In Fig. 2, all the generators are coupled directly unto the ac-bus and so supply the AC load directly. In this configuration, the diesel generator capacity must be sufficient for meeting the peak load.

The AC-coupled modular expandable hybrid systems model is based on inverters that implement reactive power/voltage and active power/frequency droops that have been developed for power control of inverters [18,19]. These droops are similar to conventional frequency/voltage droops in utility grids. Consequently, supervisory control just provides parameter settings for each component such as idle frequency, idle voltage, slopes of droops and basic commands. Communication and or extra cabling are not required in expandable inverter systems if the inverters themselves set their instantaneous active and reactive power. The inverter frequency which is the same as the mini-grid frequency is a function of the power output and several inverters can operate in parallel and at different locations. These variable frequency inverters allow for inter-connection of mini-grids. Whatever the circumstances, the inverters distribute the power amongst themselves perfectly. Other types of rotating generators—wind turbines, diesel generators or micro-hydro power plants can also be integrated without need for any overlying control structure. Since the mini-grid frequency depends on the power production capacity and the actual load, the frequency becomes a measure of how much generation capacity is being used, and so can be used as a signal for demand-side management and generation scheduling. Thus the mini-grid frequency can be used for load management and generation scheduling with communications through a radio frequency link, or over the powerline. The communications system allows for remote monitoring and control, tele-metering and billing [20].

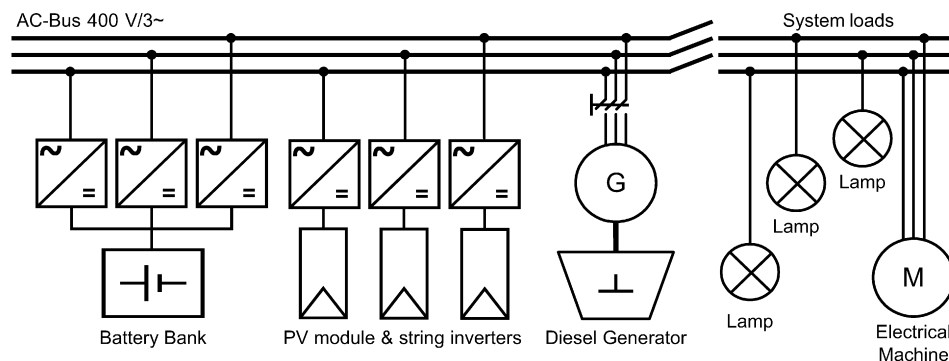


Fig. 2. Three-phase AC-coupled modular expandable hybrid system.

3. Off-grid generation options

The off-grid generation options under consideration are based on the AC-bus configuration. HOMER [21] hybrid system simulation software that can simulate optimal solutions for hybrid power systems based on the DC, AC/DC or AC-bus configuration is used in this paper. HOMER is an optimisation model, which performs many hundreds or thousands of approximate simulations in order to design the optimal hybrid system. It is simple and efficient enough to conveniently evaluate a large number of design options in search of the optimum. HOMER has been used by a number of researchers to determine an optimum hybrid energy system [22,23].

In this simulation, the hybrid power systems considered generate electricity from the combination of hydro or solar resources with a diesel or a LPG generator. Unlike fossil fuel that must be bought and transported to most remote villages, LPG can be produced locally using LPG digesters [24,25]. Battery storage is included to reduce the number of start/stop cycles of the backup diesel generator in the hybrid systems simulated as proposed in Ref. [26]. It has been proven in Ref. [9] that PV hybrid systems are more economical than standalone systems, thus the following hybrid systems were considered in this simulation:

- Micro hydro/LPG generator/battery systems (S1).
- Micro hydro/diesel generator/battery systems (S2).
- PV/LPG generator/battery systems (S3).
- PV/diesel generator/battery systems (S4).

The hydro hybrid system under consideration is based on the successful implementation of a community micro-hydro project in Kenya in 2001. The project uses an 18 kW turbine and generates about 14 kW of electricity at 415V–50 Hz. The turbine exploits a nominal flow of 200 l/s through a head of 13 m, a penstock of length 18.5 m and a steel pipe diameter of 300 mm. The project was realised for the connection of 300 households (3000 people) at a cost of 3495 US\$/kW [27]. The simulation of PV hybrid systems is motivated by two PV-powered single-phase mini-grids (220V, 50 Hz) each of length 7 km in two Senegalese villages with 2000 inhabitants [28]. A careful investigation of the energy needs of the two villages each located at a little more than 15 km from the national grid had resulted in the design of two stand alone PV systems rated 21.5 and 19.8 kWp for an estimated village daily energy consumption of 70.7 and 60 kWh with an average of 0.37 kWh/day per household [28]. These systems were installed in 1989/90

and upgraded to 26.64 and 20 kWp in 1995/1996 to solve problems of overloading.

4. Simulation data

Single-wire earth return grid extension was also considered as an alternative to these hybrid systems. From data on grid extension costs in developing countries, the extension cost for single-wire earth return systems was evaluated at 5000 €/km [29]. The annual operation and maintenance costs were computed as 2.5% of the grid extension cost giving 125 €/yr/km while the grid power price was set to the local grid price of 0.1 €/kWh.

For this simulation, the solar resource of Garoua (9°20' N, 13°23' E) a location in the North Province of Cameroon was selected for remote villages in northern parts of the country. This solar resource was selected as it has been recently used in Ref. [10] for the modelling of solar hybrid systems for far north Cameroon. The monthly solar insolation obtainable at Garoua is in the range of 4.6–6.1 kWh/m²/day. The hydro resource for this simulation was obtained by scaling the annual average flow of River Mungo which flows through the South West and Littoral provinces [30].

The monthly solar and hydro resource patterns used in this simulation are shown in Table 1 and Figs. 3–4. The annual solar insolation of Garoua evaluated at 5.5 kWh/m²/day falls within the range of solar insolation levels that have been used in the deployment of thousands of PV systems worldwide for the supply of electricity to remote communities [26]. The solar insolation of Garoua is higher during the dry months between January to May and October to December implying that a fuel generator would be mostly used during the wet months when the solar insolation is lower. The water flow pattern shown in Fig. 4 confirms the situation that affects hydroelectric generation in the south of the country. A fuel generator is therefore also required during the dry months in the south.

These resource patterns were considered for the generation of electricity that could meet the energy needs of a typical village. To do this the power and energy demand data of 19 grid-connected households in a village in Uganda located near the capital city Kampala was used [31]. The data are shown per phase in Tables 2 and 3 with the day segmented into four periods. In Table 2, *L1*, *L2*, *L3*, and *L* represent the power demand per phase and the total power demand of the households on a three-phase rural distribution transformer. Similarly in Table 3, *E1*, *E2*, *E3* and *E* represent the energy demand per phase and the

Table 1
Solar and hydro resources of Garoua and river Mungo

Resource	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Ann
Solar (kWh/m ² /d)	4.6	5.9	6.1	6.2	6.1	5.6	5.0	5.3	5.6	5.6	5.4	5.2	5.5
Hydro (m ³ /s)	46	42	55	88	114	170	263	321	343	276	162	75	163

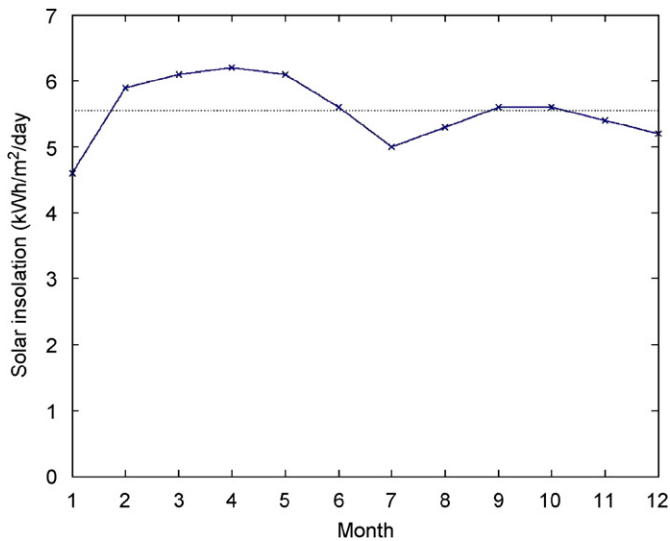


Fig. 3. Monthly solar insolation of Garoua located in North Cameroon.

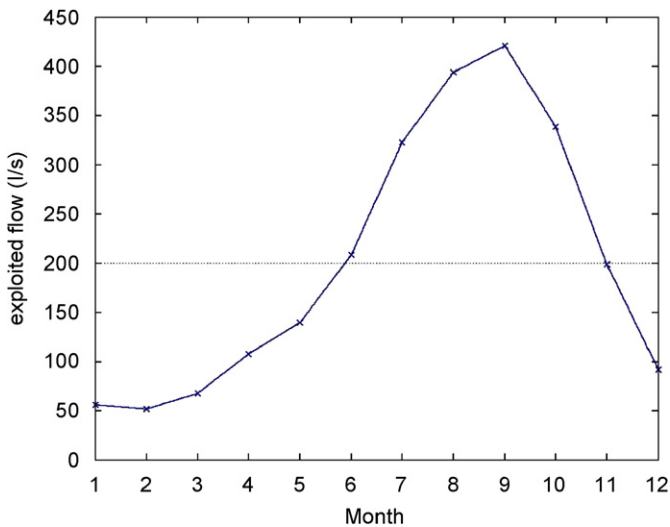


Fig. 4. Exploited flow derived from the flow of River Mungo.

Table 2
Power demand per phase for 19 grid-connected households in Najjeera village, Uganda [31]

Load	Period			
	00–06	06–09	09–19	19–24
L1 (kW)	1.12	1.70	0.62	2.92
L2 (kW)	0.86	1.90	2.19	4.20
L3 (kW)	1.15	1.04	0.16	2.64
L (kW)	3.13	4.64	2.97	9.76

total energy demand of households on the transformer. It is observed in Table 3 that the daily energy demand of the 19 grid-connected households is 110 kWh.

The data in Table 2 show that there is a morning peak between 6 and 9 am and an evening peak between 7 pm and midnight. Morning and evening peaks have also been

Table 3

Energy demand per phase for 19 grid-connected households in Najjeera village, Uganda [31]

Energy	Period			
	00–06	06–09	09–19	19–24
E1 (kWh)	6.64	5.03	6.16	14.50
E2 (kWh)	5.05	5.54	21.64	20.83
E3 (kWh)	6.81	3.06	1.59	13.12
E (kWh)	18.50	13.63	29.39	48.45

observed in the load profile that has been constructed for urban households in Cameroon [32]. Consequently, these data were used as a typical village load profile in the simulation of off-grid electricity generation options for remote villages. Using these load data as primary load with an hourly and daily noise of 5% HOMER generated an annual load profile with a daily energy demand of 111 kWh/day and a peak power demand of 12 kW. Thus a 15 kW fuel (LPG/diesel) generator was selected for meeting the peak load and possible growth in energy demand. In the absence of a fuel generator, the battery inverters must be of sufficient capacity to meet the peak power demand. The 1-week load profile generated with HOMER is shown in Fig. 5.

In order that HOMER should compute the levelized costs of energy for the different renewable energy options under consideration, current financial information for power components in the different options must be specified. From the cost of a community micro-hydro project with a rated turbine output of 18 kW in Kenya [27], the capital, replacement and operation and maintenance costs of the hydro system were estimated at 10,000€, 5000€ and 1000€/year, respectively. In Ref. [9], the costs of generators rated 5, 25, 50, 75 and 100 kW are stated as being equal to 533, 325, 220, 183 and 160 €/kW respectively. Using these generator sizes and prices the cost of a 15 kW diesel generator on European markets was estimated using linear interpolation at 410 €/kW. Applying a 40% price increase in local markets, the cost of a 15 kW diesel or LPG generator was evaluated at 8610€. From the local cost of 12 kg low pressure gas bottles and a 16% increase for transportation to remote villages, the costs of LPG was computed as 0.7 €/m³. Similarly the cost of petrol was calculated from the prices in urban areas as 1 €/l.

Using prices from retailers for European markets, the capital and replacement cost of a 3 kWp PV array and string inverter [9] was estimated at 15,000€ while that of a 3.3 kW bidirectional inverter was evaluated with a 40% increase at 3080€. Similarly, the capital and replacement costs of 2 V Hoppecke 12 OPzS 1500 vented lead-acid, tubular plate, deep cycle batteries rated 3 kWh were estimated with a 40% increase at 1020€ from the costs of Exide OPvS batteries. The operation and maintenance cost of the 3 kWp PV array and string inverter as well

as that of a 2 V deep cycle battery were evaluated at 0.05% of their capital costs [9] at approximately 8 and 51 €/yr. The round trip efficiency of the selected battery is 82% and its minimum state of charge is 30%. The efficiency of the bidirectional inverter is 90% while that of the PV-string inverter is assumed to be 90% as in Ref. [9]. The financial data used in this simulation are shown in Table 4.

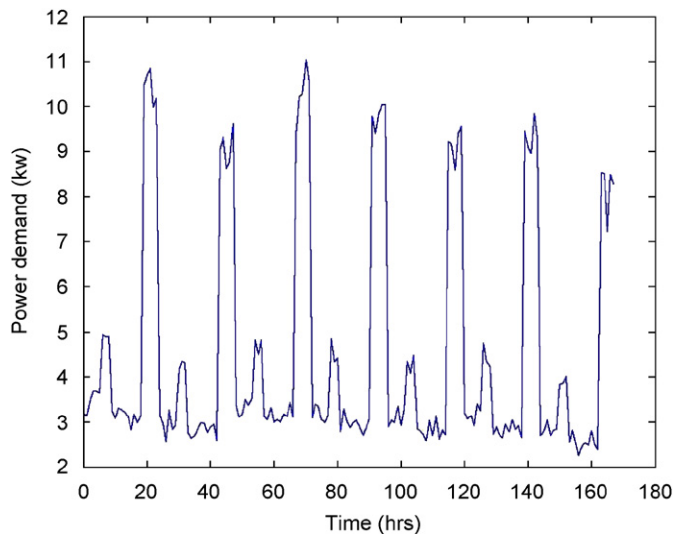


Fig. 5. 168-h load profile extracted from the annual hourly load data generated with HOMER.

Table 4
Financial data for simulated options

Description of system parameters	Value
3 kW PV array and string inverter capital and replacement costs (€)	15,000
3.3 kW battery inverter capital and replacement costs (€)	3080
2 V Hoppecke 12 OPzS 1500 Ah battery capital and replacement costs (€)	1020
Operation and maintenance cost of 3 kWh battery (€/yr)	50
Operation and maintenance cost of 3 kW PV array (€/yr)	8
15 kW fuel generator capital and replacement costs (€)	8610
Fuel generator operation and maintenance costs (€/h)	0.80
Capital costs of 18 kW turbine + generator (€)	10,000
Replacement cost of 18 Kw turbine + generator (€)	5000
Micro-hydro operation and maintenance costs (€/yr)	1000
Micro-hydro fixed system cost	8000
Micro-hydro system fixed operation and maintenance costs	1000
Petrol costs (€/l)	1.00
LPG costs (€/m ³)	0.70
PV fixed system capital cost (€)	1000
PV system fixed capital operation and maintenance costs (€/yr)	0
System shortage penalty (€/kWh)	0
Carbon dioxide emissions penalty (€/t)	1.5
Annual real interest rates (%)	6
Project lifetime (yrs)	25
Diesel/LPG generator lifetime or operating hours (hrs)	30,000
Cost of grid extension (€/km)	5000
Operation and maintenance cost of grid line (€/yr/km)	125
Grid power price in Cameroon (€/kWh)	0.10

5. Hybrid system optimisation procedure

As a consequence of the 24 V DC-input voltage of the selected bidirectional inverter, batteries that were considered in the battery module were 12, 24 and 36. In the converter module, total power converter capacity ratings of 3.3, 6.6 and 9.9 kW were selected. In the PV generator module, five values for PV capacity 9, 12, 15, 18, 21 kW were considered. In the hydro module, the minimum and maximum exploitable flows for the micro-hydro scheme were set to 50% and 100% of the nominal flow, respectively. Using data of the 18 kW cross-flow turbine that was used in Kenya [27], the available head was set to 13 m, the design flow to 200 l/s and the turbine-generator efficiency was set to 55% to obtain an output of 14 kW. In the head loss module, the steel pipe diameter was set to 300 mm and the length of the pipe to 18.5 m and the head loss was evaluated as 2.82%. Since the rating of the fuel/LPG generator is close to 12 kW, the minimum load ratio was set to 70% in the generator modules as adopted in [10].

In the generator control module, cycle charging and a state set point of charge of 50% was selected. In the constraints module the maximum annual capacity shortage was set to 0% while the minimum renewable fraction was set to 60%. This renewable energy fraction was selected since renewable energy fractions greater than 60% have been obtained in PV hybrid systems that have been simulated for villages in Mekong countries [9]. An hourly operating reserve of 10% was specified to ensure that the system can cope with a 10% increase in the hourly primary load. Other relevant parameters in the economics, generator control, carbon emissions, and constraints modules were also specified.

The effect of stream flow on the levelised energy costs of hydro hybrid systems was simulated with six values of scaled annual flow of 100 l, 120 l, 140 l, 160 l, 180 l and 200 l/s. The effect of changes in the capital cost of PV generator on the levelised cost of energy of PV hybrid systems was also investigated by using six sensitivity values 1, 1.2, 1.4, 1.6, 1.8 and 2 in the PV generator module.

6. Results

The results presented here were obtained in two separate simulations involving hydro or PV hybrid systems. The results obtained with PV capital costs sensitivity set to 1.4 and a scaled annual flow of 200 l/s are shown in Table 5. In Table 5, PV represents PV array power, PH hydro generator power, B battery storage, G_1 diesel generator power, G_2 LPG generator power, BC bidirectional converter power, C capital costs of power system, NPC net production costs, COE cost of energy, RF renewable energy fraction, f litres of fuel, v volume of LPG, h_1 diesel generator hours, h_2 LPG generator hours, E percentage of excess energy, d breakeven grid distance.

It is observed in Table 5, that the cost of the energy offered by option S1 is cheaper than option S2 for villages

Table 5

AC-bus configurations, energy costs, breakeven grid distances with PV capital multiplier equal to 1.4 and exploited flow of 200l/s

Option	PV (kW)	PH (kW)	G_1 (kW)	G_2 (kW)	B (kWh)	BC (kW)	C (€)	NPC (€)	COE (€/kWh)	RF (%)	f (l)	v (m ³)	h_1 [h]	h_2 [h]	E [%]	d [km]
S1	0	14	0	15	36	6.6	45010	153234	0.296	82.1	0	5452	0	1260	51	15.4
S2	0	14	15	0	36	6.6	45010	174226	0.336	82.1	5452	0	1260	0	51	18.6
S3	18	0	0	15	72	9.9	169330	298376	0.576	60.7	0	7408	0	1678	16	37.4
S4	18	0	15	0	72	9.9	169330	326898	0.631	60.7	7408	0	1678	0	16	41.7

in the southern parts of the country that possess the exploited flow shown in Fig. 4. In Table 5, option S3 is more economical than option S4 for villages in the northern parts of the country that have at least the solar insolation shown in Fig. 3. The cost of energy for the hybrid systems shown in Table 5, could be reduced if LPG is generated locally from animal or human waste. LPG production using existing technology [25] could also be used for domestic thermal needs.

The variation of the cost of energy and breakeven grid distances with changes in the annual flow exploited in options S1 and S2 are shown in Table 6. The variation of the cost of energy and breakeven grid distances with changes in PV capital cost multiplier in options S3 and S4 are shown in Table 7. It is observed in Table 6 that an increase in the scaled annual flow exploited by the hydro installation reduces the cost of energy and the grid distance at which these systems become more economical than grid extension. In Table 7, an increase in PV capital cost multiplier increases the cost of energy and the grid distance at which PV hybrid systems become more economical than grid extension. In spite of the wide variation of PV capital cost multiplier the cost of energy obtained falls within the range of values (0.69–0.55 €/kWh) evaluated for PV hybrid systems for villages in Mekong countries [9] when power conditioning components are imported from Europe. The variations of the cost of energy and breakeven grid distances with changes in the scaled annual flow exploited by the hydro hybrid system are shown in Figs. 6–7. The variations of the cost of energy and breakeven grid distances with changes in PV capital costs are presented in Figs. 8–9.

It is obvious from Figs. 6 and 8, that option S1 is the cheapest option for villages that have streams with an annual flow of 200l/s while option S3 is the best option for villages with an annual solar insolation of at least 5.55 kWh/m²/day. For an annual flow of 200l/s and a PV capital cost multiplier value of 1.4, the energy costs and the breakeven grid distances for feasible options were found to be:

- 0.296 €/kWh and 15.4 km, respectively, for option S1;
- 0.576 €/kWh and 37.4 km, respectively, for option S3.

These off-grid options compete favourably with grid extension and so could be used in the current National Energy Action Plan for the provision of energy services to

Table 6

Cost of energy and breakeven grid distances for hydro hybrid options

Flow (l/s)	100	120	140	160	180	200
COE (€/kWh)	S1 0.415 S2 0.486	0.376 0.437	0.374 0.435	0.335 0.386	0.334 0.385	0.296 0.336
d (km)	S1 24.7 S2 30.3	21.7 26.4	21.5 26.3	18.4 22.4	18.4 22.3	15.4 18.6

Table 7

Cost of energy and breakeven grid distances for PV hybrid options

x	1	1.2	1.4	1.6	1.8	2
COE (€/kWh)	S3 0.507 S4 0.562	0.541 0.596	0.576 0.631	0.611 0.666	0.646 0.701	0.68 0.735
d (km)	S3 31.4 S4 36.2	34.6 39	37.4 41.7	40.1 44.4	42.8 47.2	45.6 49.9

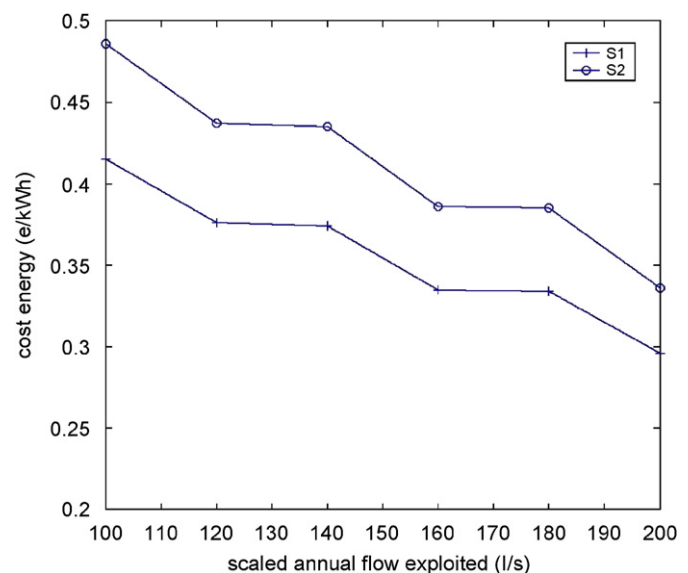


Fig. 6. Levelised cost of energy for hydro hybrid options for typical village load.

remote villages, schools, rural health centres and rural businesses that are often located at distances outside the breakeven distances that have been computed. These off-grid options can be used to reduce the number of villages that currently lack grid power supplies [3] and also the national poverty level of 40.2 [1].

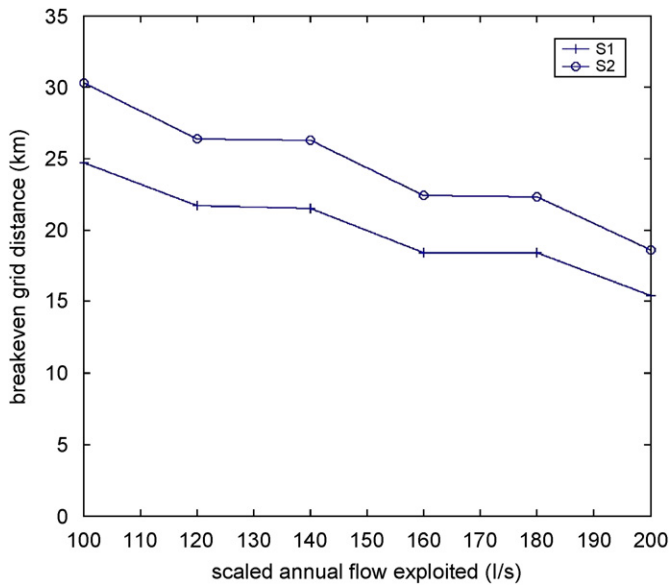


Fig. 7. Breakeven grid distance for hydro hybrid options for typical village load.

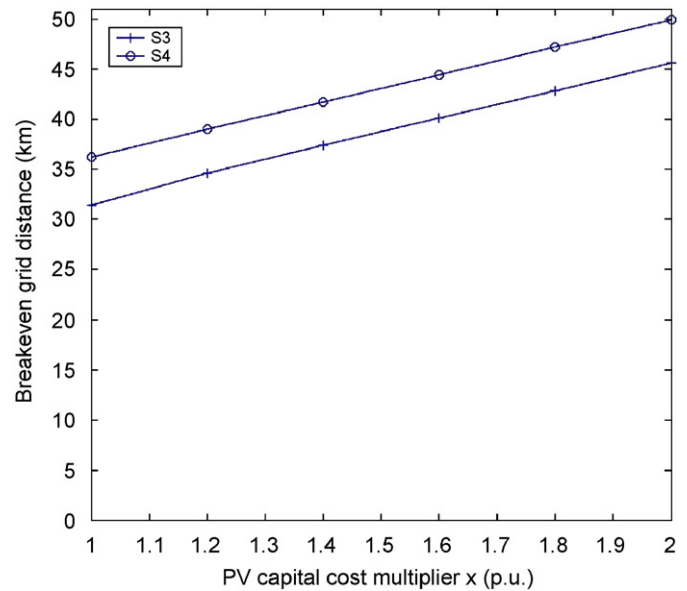


Fig. 9. Breakeven grid distance for PV hybrid options for typical village load.

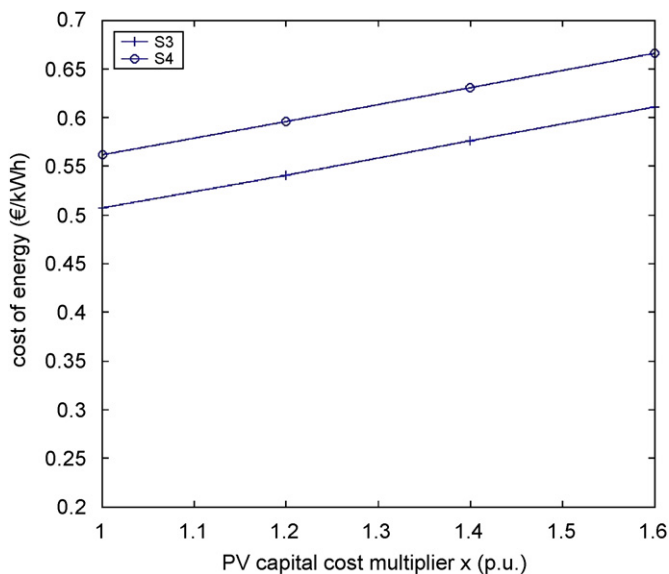


Fig. 8. Cost of energy for PV hybrid options for typical village load.

7. Conclusion

Off-grid electricity generation options for the electrification of remote villages in Cameroon have been simulated with a typical load profile with an energy demand of 111 kWh/day and 12 kWp. Micro-hydro/LPG/battery systems were found to be the best supply options for remote villages located in the southern parts of Cameroon, while PV/LPG/battery systems proved to be the best option for villages in the northern parts of Cameroon.

The results of this simulation showed that for a fuel cost of 1 €/l, a 14 kW micro-hydro plant with a 15 kW LPG generator and 36 kWh battery storage produced electricity at a cost of 0.296 €/kWh for remote communities with an

annual flow of 200 l/s, a head of 13 m, a penstock of length 18.5 m and diameter of 300 mm. It was also found that an 18 kWp PV generator with a 15 kW LPG generator and 72 kWh battery storage produced electricity at a cost of 0.576 €/kWh for remote areas with an average annual insolation of at least 5.55 kWh/m²/day.

These results showed that off-grid options based on renewable energy resources could be a suitable alternative for rural electrification in the low power range (10–50 kW) in Cameroon. Thus electricity needs of most remote villages in Cameroon can be met with these options while biogas technology could be used to reduce the cost of energy with excess biogas serving domestic thermal needs since this has been implemented elsewhere [24]. The production of biogas from potential sources could be based on the renewable energy technology model that has been proposed for the empowerment of rural people in Bangladesh [25].

In order to deploy these off-grid systems in Cameroon, renewable energy databases for all regions need to be developed. In addition, there is also the need to study rural energy needs as well as their socio-economic status. This site-specific information is important for the survival of systems that would be sized and installed in remote communities in Cameroon.

References

- [1] République du Cameroun. Conditions de vie des ménages et profile de pauvreté à l'Extrême-Nord Cameroun en. 2001. Direction de la Statistique et de la Comptabilité Nationale, Ministère de l'Economie et des Finances, Yaoundé, 2002.
- [2] Tchouate HPM. Contribution des Energies Renouvelables au Développement Durable du Secteur de Electricité: Le Cas du Cameroun, Thèse de Doctorat. Belgique: Université Catholique de Louvain; 2003.

- [3] Mihamle JD. Cameroon lack of electricity: a hostage situation. African News Bulletin–Bulletin d'Information Africaine, Supplement Issue 2002;445.
- [4] Kenfack F, Guinet A, Ngundam JM. Investment planning for electricity generation expansion in a hydro-dominated environment. *Int J Energy Res* 2001;25:927–37.
- [5] Ngundam JM, Kenfack F, Tatiése TT. Optimal scheduling of large-scale hydrothermal power systems using Lagrangian technique. *Electr Power Energy Syst* 2000;22:237–45.
- [6] Dung TQ, Anisuzzaman M, Kumar S. Demonstration of multi-purpose battery charging station for rural electrification. *Renew Energy* 2003;28:2367–78.
- [7] Gustavsson M, Ellegard A. The impact of solar home systems on rural households. Experiences from Nyimba energy services company in Zambia. *Renew Energy* 2004;29:1059–72.
- [8] Bryne J, Shen B, Wallace W. The economics of sustainable energy for rural development: a study of renewable energy in China. *Energy Policy* 1998;26(1):45–54.
- [9] Nippon K. PV hybrid systems for rural electrification in the Mekong Countries. PhD Thesis, University of Kassel, Germany, 2005.
- [10] Nfah EM, Ngundam JM, Tchinda R. Modelling of solar/diesel/battery hybrid power systems for far North Cameroon. *Renew Energy* 2007;32(5):832–44.
- [11] Oparaku OU. Rural area power supply in Nigeria: a cost comparison of the photovoltaic, diesel/gasoline generator and grid utility options. *Renew Energy* 2003;28:2089–98.
- [12] Maher P, Smith NPA, Williams AA. Assessment of pico-hydro as an option for off-grid electrification in Kenya. *Renew Energy* 2003;28:1357–69.
- [13] Tchinda R, Tsochounie CJH, Kaptoum E. Wind characteristics and energy potentialities of some selected sites in Northern Cameroon. *Botswana J Technol* 2004;58–63.
- [14] Meinhardt M, Rothert M, Wollny M, Engler A. Pure AC-Coupling–The concept for Simplified Design of Scalable PV-Hybrid Systems Using Voltage/Frequency Statics Controlled Battery Inverters. *PVSEC-14*; 2004. [26–30 January].
- [15] Ross MMD. Comparison of AC, DC, and AC/DC bus configurations for PV hybrid systems. CAMNET Energy Technology Centre–Varennnes, 2004.
- [16] Cramer G, Reekers J, Rothert M, Wollny M. The future of village electrification—more than two years of experience with AC-coupled hybrid systems. In: Second European PV-Hybrid Mini-grid Conference, Kassel, Germany, 25–26 September 2003.
- [17] Strauss P, Engler A. AC coupled pv hybrid systems and micro grid-start of the art and future trends. In: Third World Conference on PV Energy Conversion, Osaka, Japan, 11–18 May 2003.
- [18] Engler A. Applicability of droops in low voltage grids. *Int. J. DER ISSN1614-7138*, vol. 1 no. 1, Germany, Kassel: Technology and Science Publisher, 2005.
- [19] Engler A, Hardt C, Bechtel N, Rothert M. Next generation of ac coupled hybrid systems-3 phase parallel operation of grid forming battery inverters. In: Second European PV-Hybrid Mini-grid Conference, Kassel, Germany, 25–26 September 2003.
- [20] Schmid J, Strauss P, Schweider-Reis P. PV Mini-grids for rural development and economic growth. *Renew Energy World* 2003;6(4):192–9 [July–August].
- [21] National Renewable Energy Laboratory. Hybrid Optimization Model for Electric Renewables (HOMER), <<http://www.nrel.gov/international/tools/HOMER/homer.html>>, 2005.
- [22] Khan MJ, Iqbal MT. Pre-feasibility study of stand-alone hybrid energy systems for applications in Newfoundland. *Renew Energy* 2005;30(6):835–54.
- [23] Blanco G, Manwell JF, McGowan JG. A feasibility study for wind/hybrid power system applications for New England islands. In: AWEA Annual Conference on Wind Power 2002.
- [24] Cawood W. Maphephetheni-renewables at work in a South African rural development programme. *World Renew Energy Network* 2001 [May–June].
- [25] Biswas WK, Bryce P, Diesendorf M. Model for empowering rural poor through renewable technologies in Bangladesh. *Environ Sci Policy* 2001;4:333–44.
- [26] Elhadidy MA. Performance evaluation of hybrid (wind/solar/diesel) power systems. *Renew Energy* 2002;18:401–13.
- [27] Theuri D. A case study on the community micro hydropower project in Tungu-Kabiri, Kenya. In: Workshop on Delivery Mechanisms for Rural Electrification, Bagamoyo, Tanzania, 2003.
- [28] Koita Z, Dahouenon AM. Diaoulé and Ndiébel photovoltaic plants: 6 years operating experience. *ISES Solar World Congress*, 1997.
- [29] UNDP, World Bank/ESMAP. Reducing the cost for grid extension for rural electrification. NRECA Int. Ltd, 2000.
- [30] SONEL Cameroun. Atlas du potentiel hydroélectrique du Cameroun. 1983, p. 35.
- [31] Sprei F. Characterization of power system loads in rural Uganda. Master Thesis, Lund University, Sweden, 2002.
- [32] Tatiése TT, Villeneuve P, Ngundam JM, Kenfack F. Contribution to the analysis of urban residential electrical energy demand in developing countries. *Energy* 2002;27(6):591–606.